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Kiapour et al.

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(54) **PARALLEL PREDICTION OF MULTIPLE
IMAGE ASPECTS**

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2207/20081 (2013.01); **G06T 2207/20084**

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(58) **Field of Classification Search**

None

See application file for complete search history.

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(57) **ABSTRACT**

Example embodiments that analyze images to characterize
aspects of the images rely on a same neural network to
characterize multiple aspects in parallel. Because additional
neural networks are not required for additional aspects, such
an approach scales with increased aspects.

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(51) **Int. Cl.**

G06F 16/583 (2019.01)

G06F 16/16 (2019.01)

G06N 3/045 (2023.01)

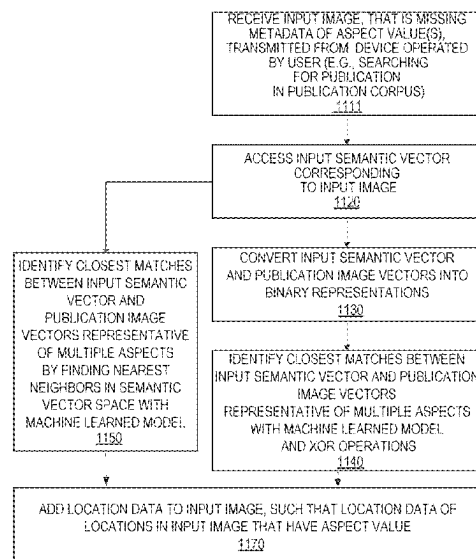
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H04L 67/10 (2022.01)
H04L 67/104 (2022.01)

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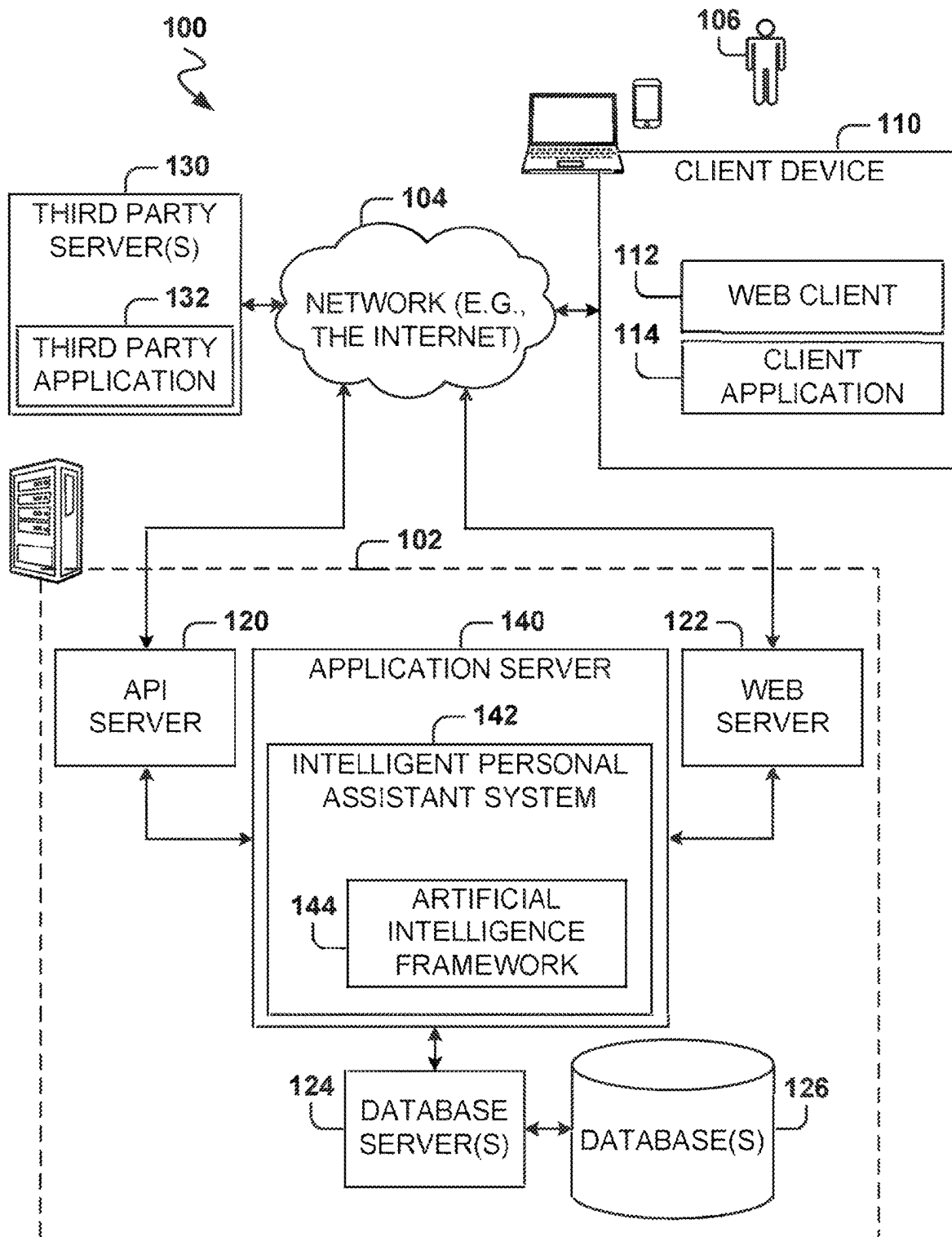


FIG. 1

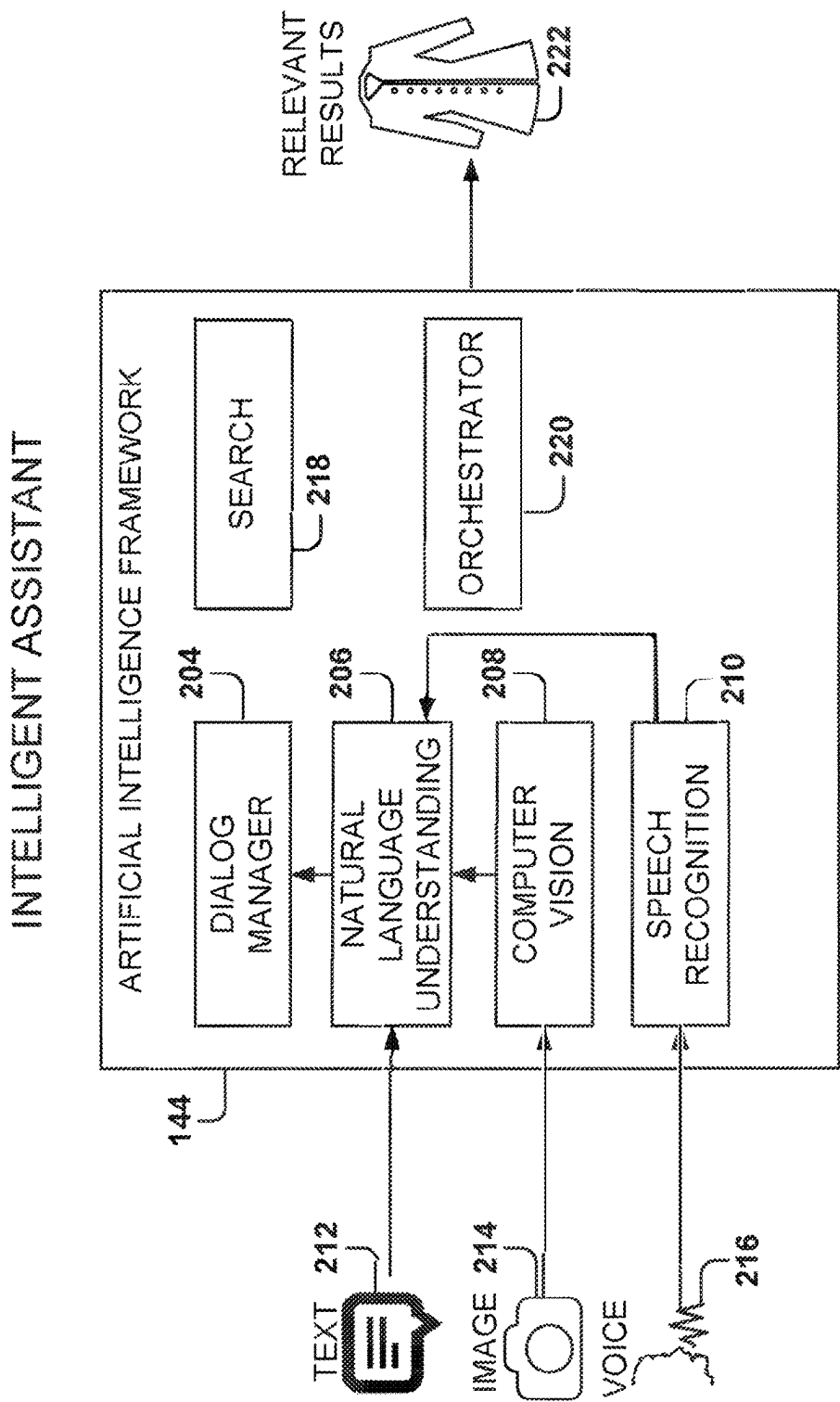


FIG. 2

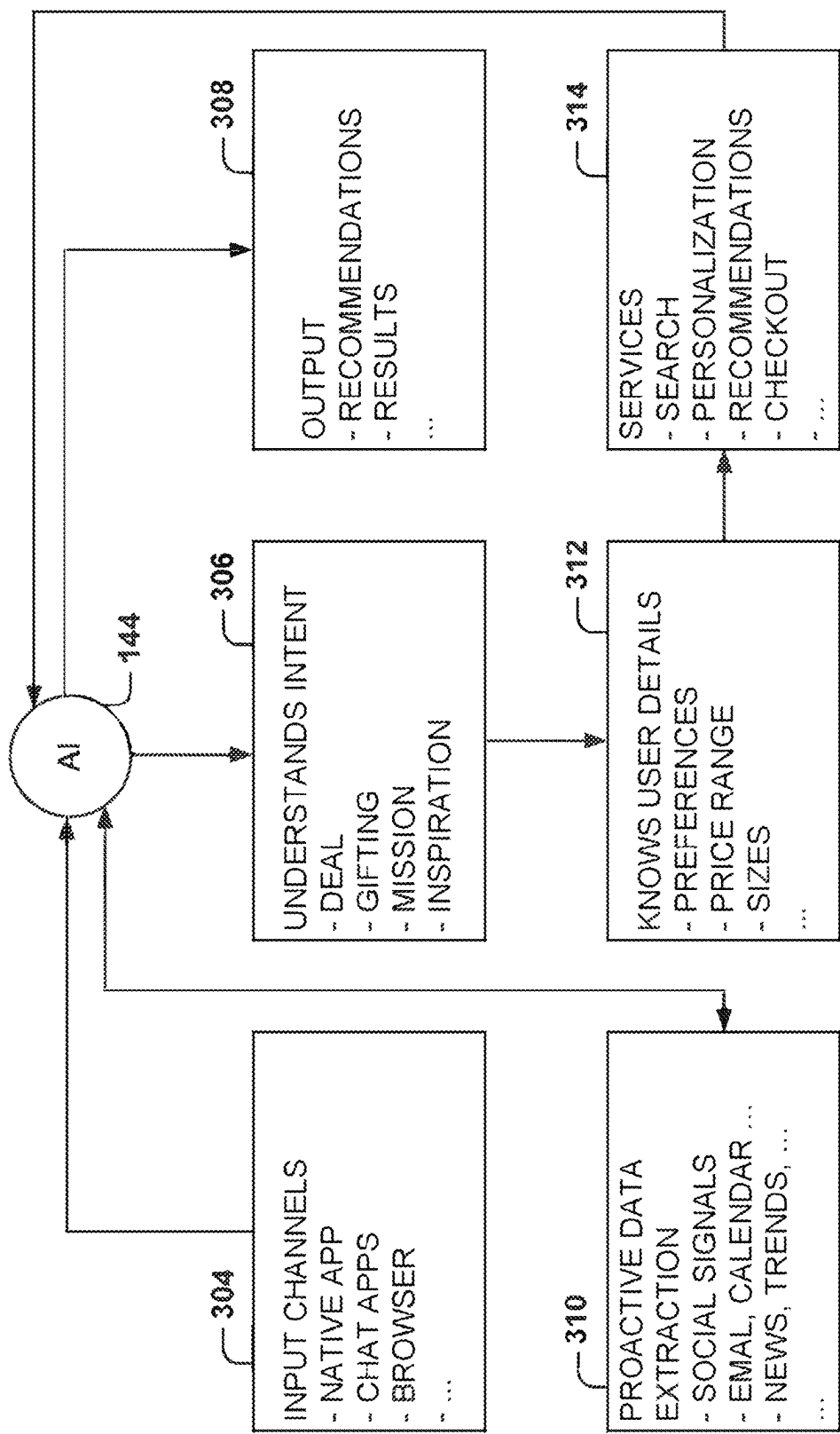


FIG. 3

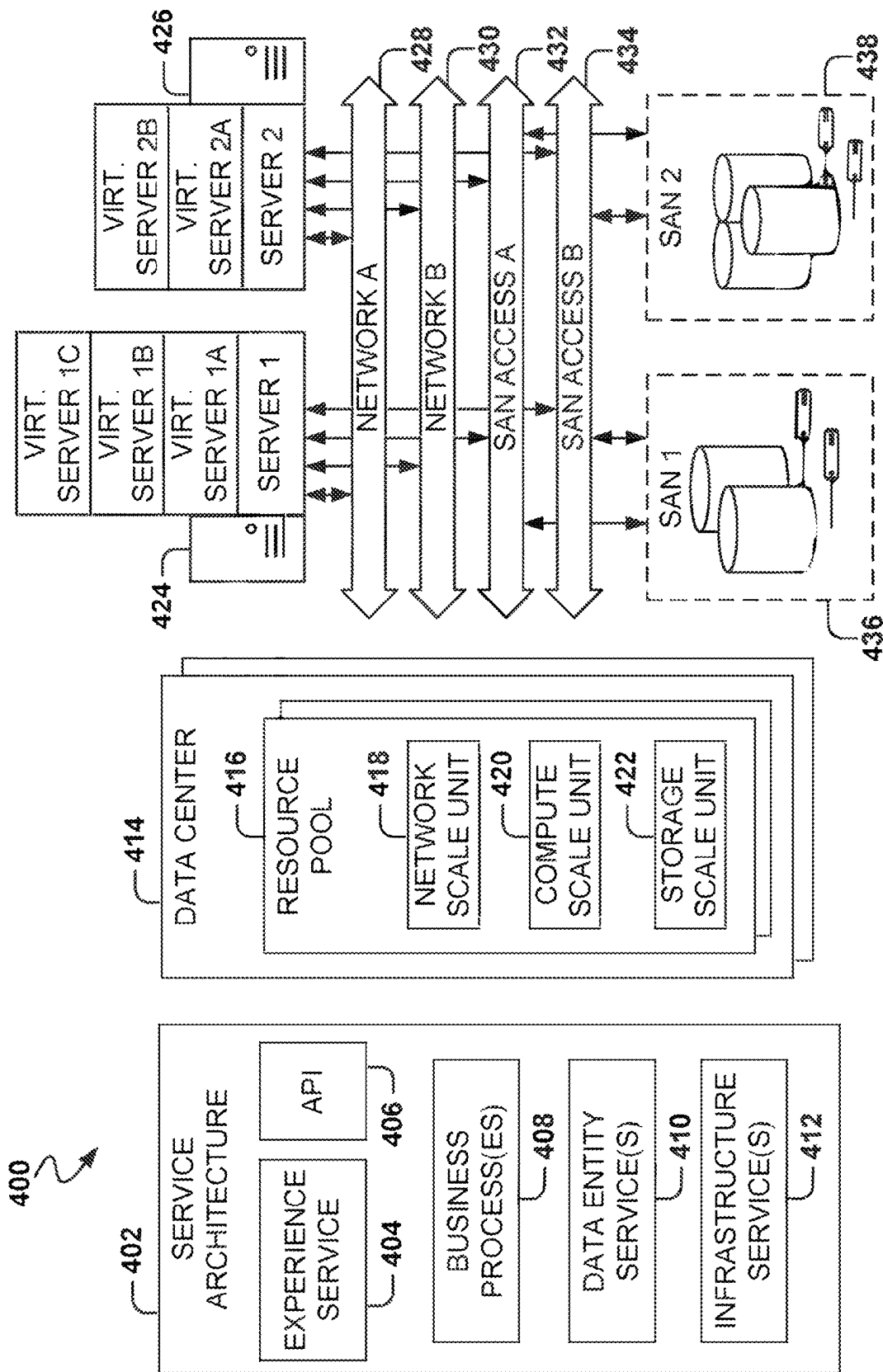


FIG. 4

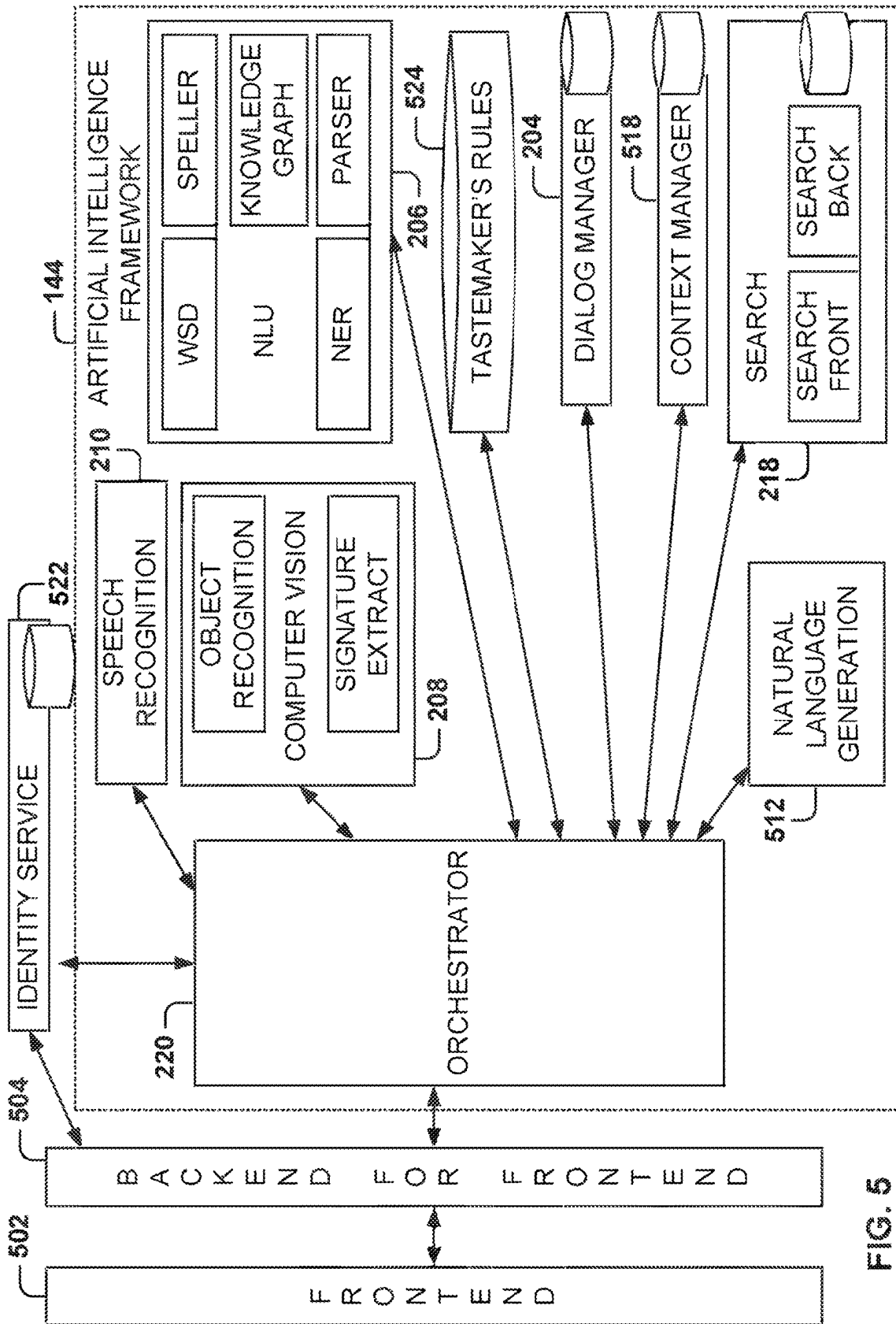


FIG. 5

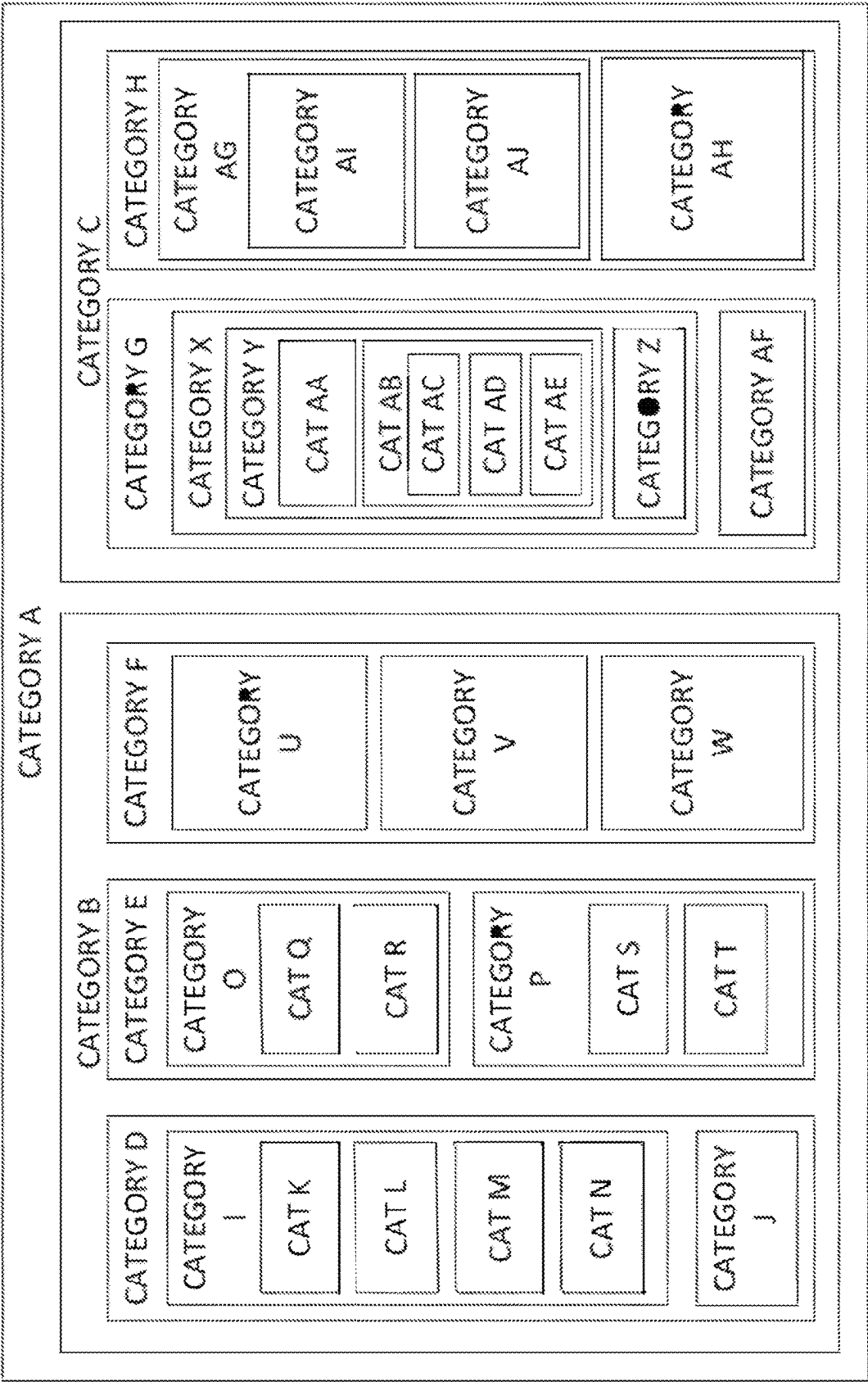


FIG. 6

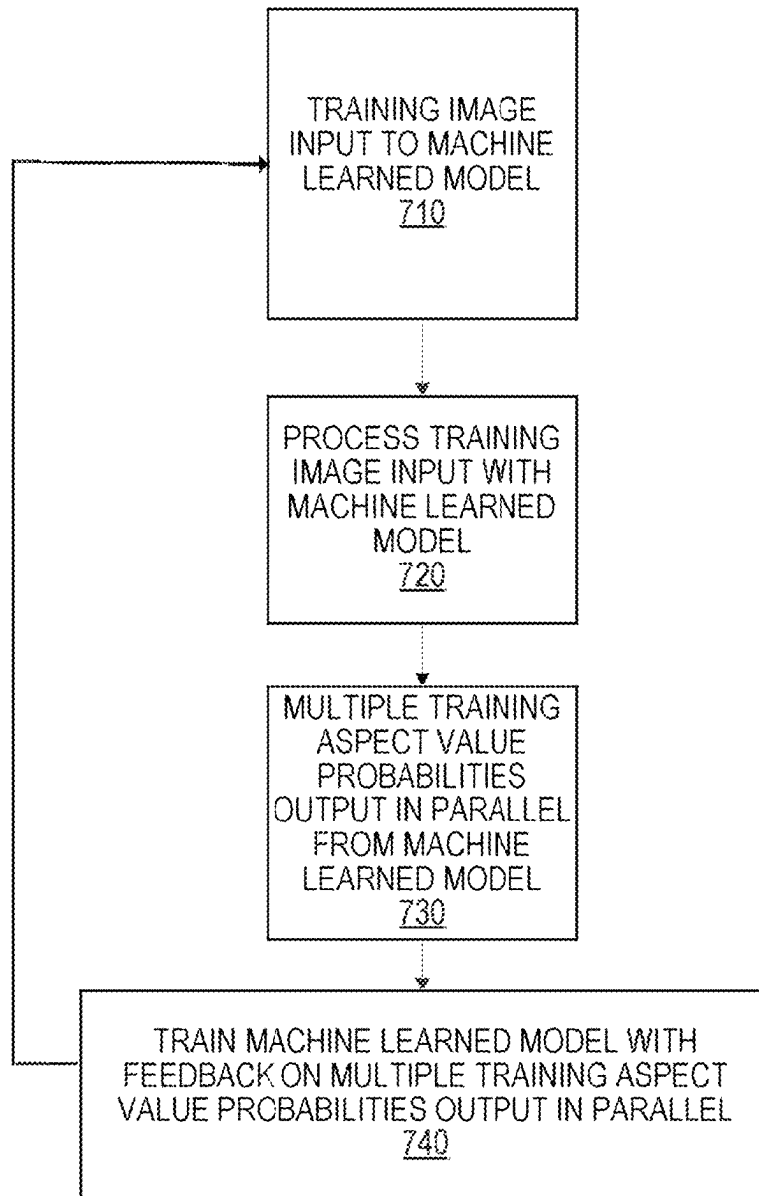


FIG. 7

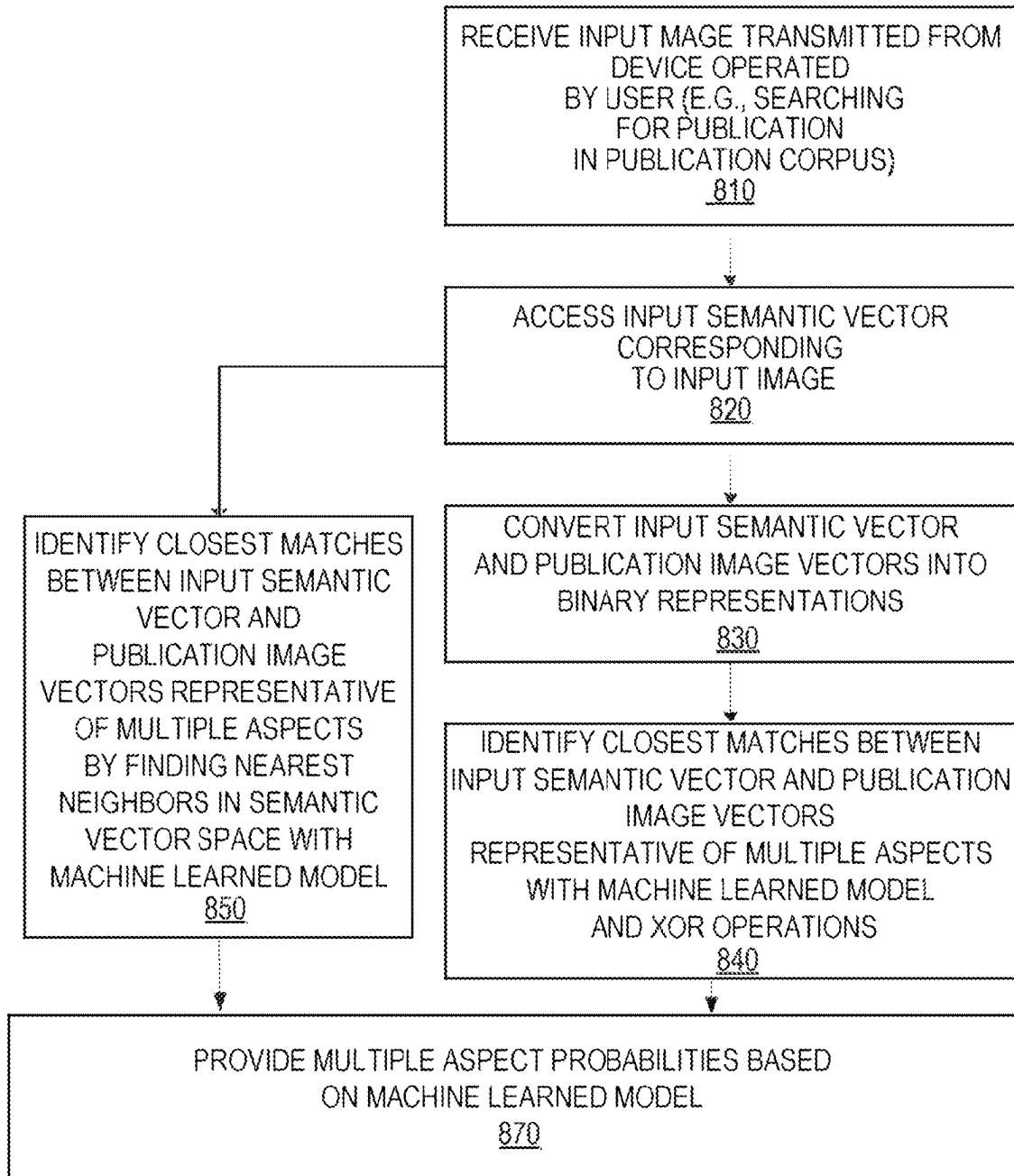


FIG. 8

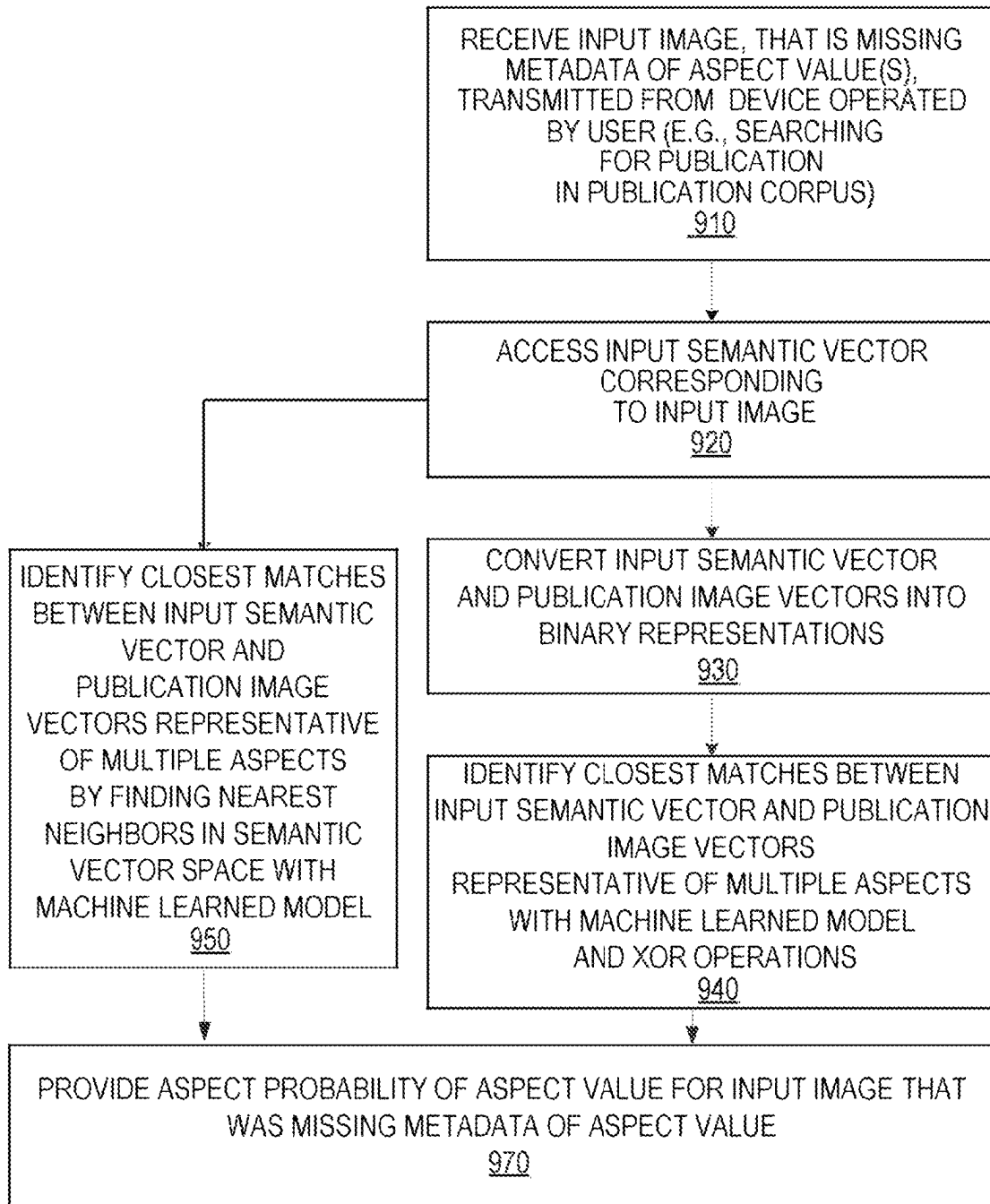


FIG. 9

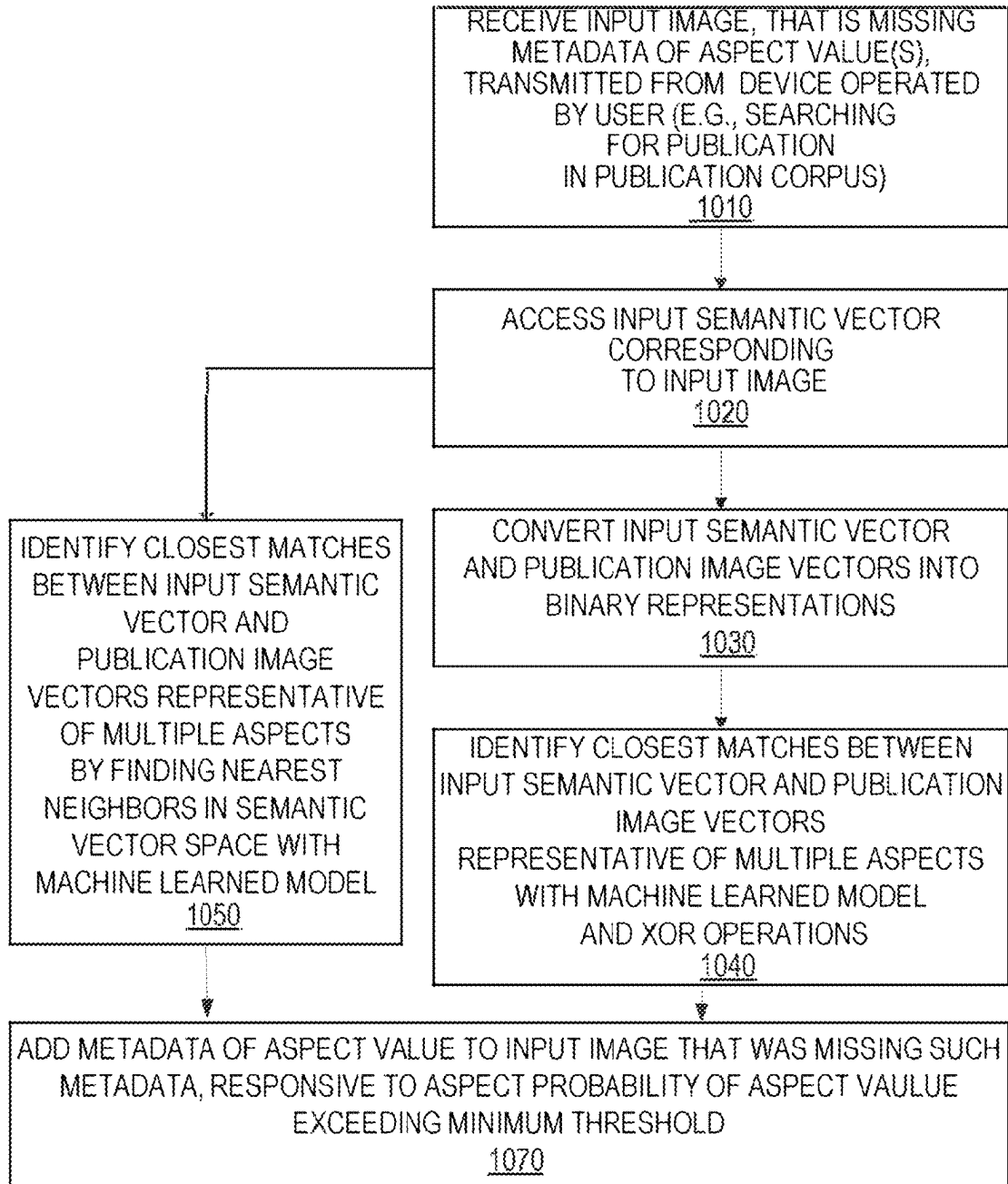


FIG. 10

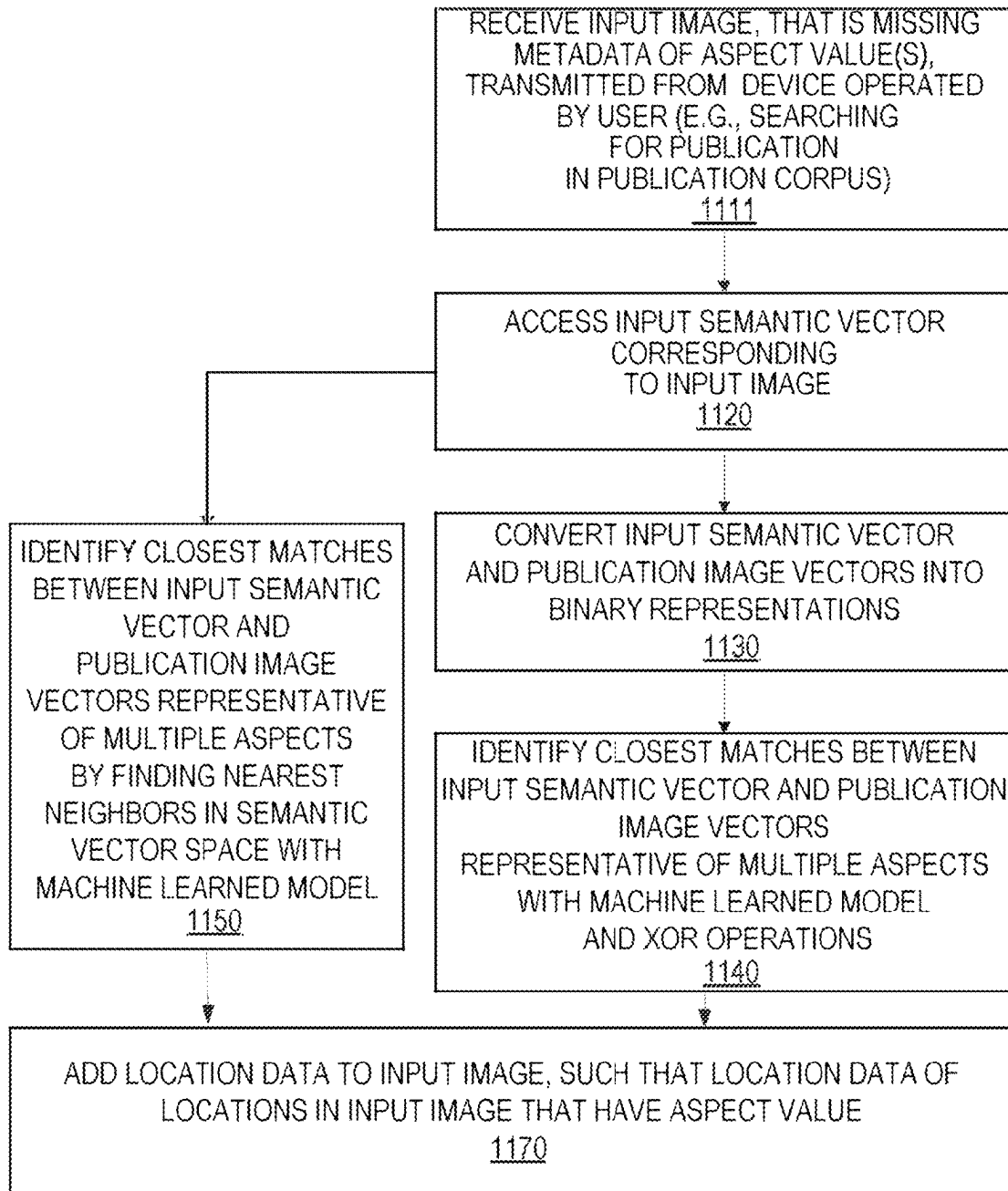


FIG. 11

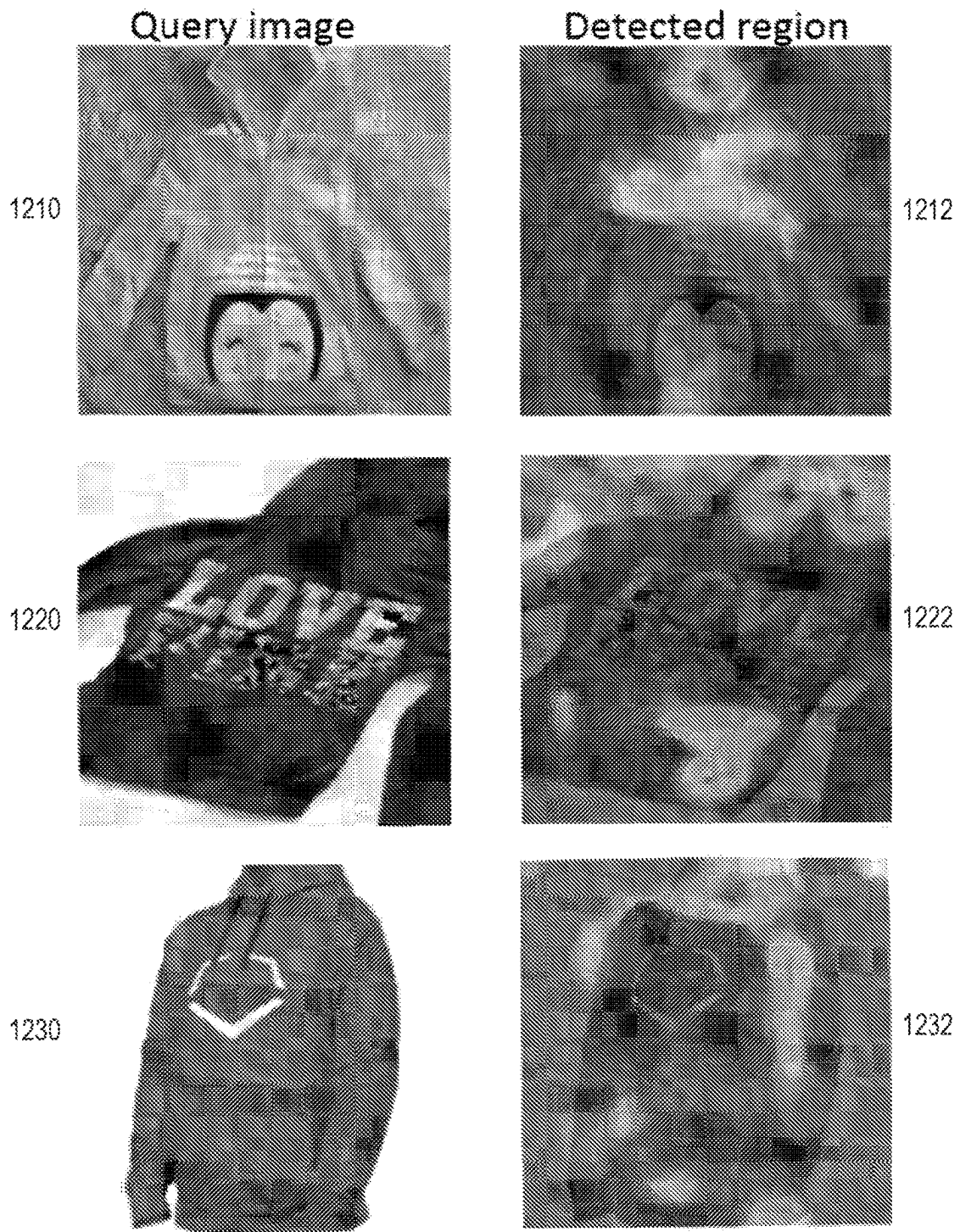


FIG. 12

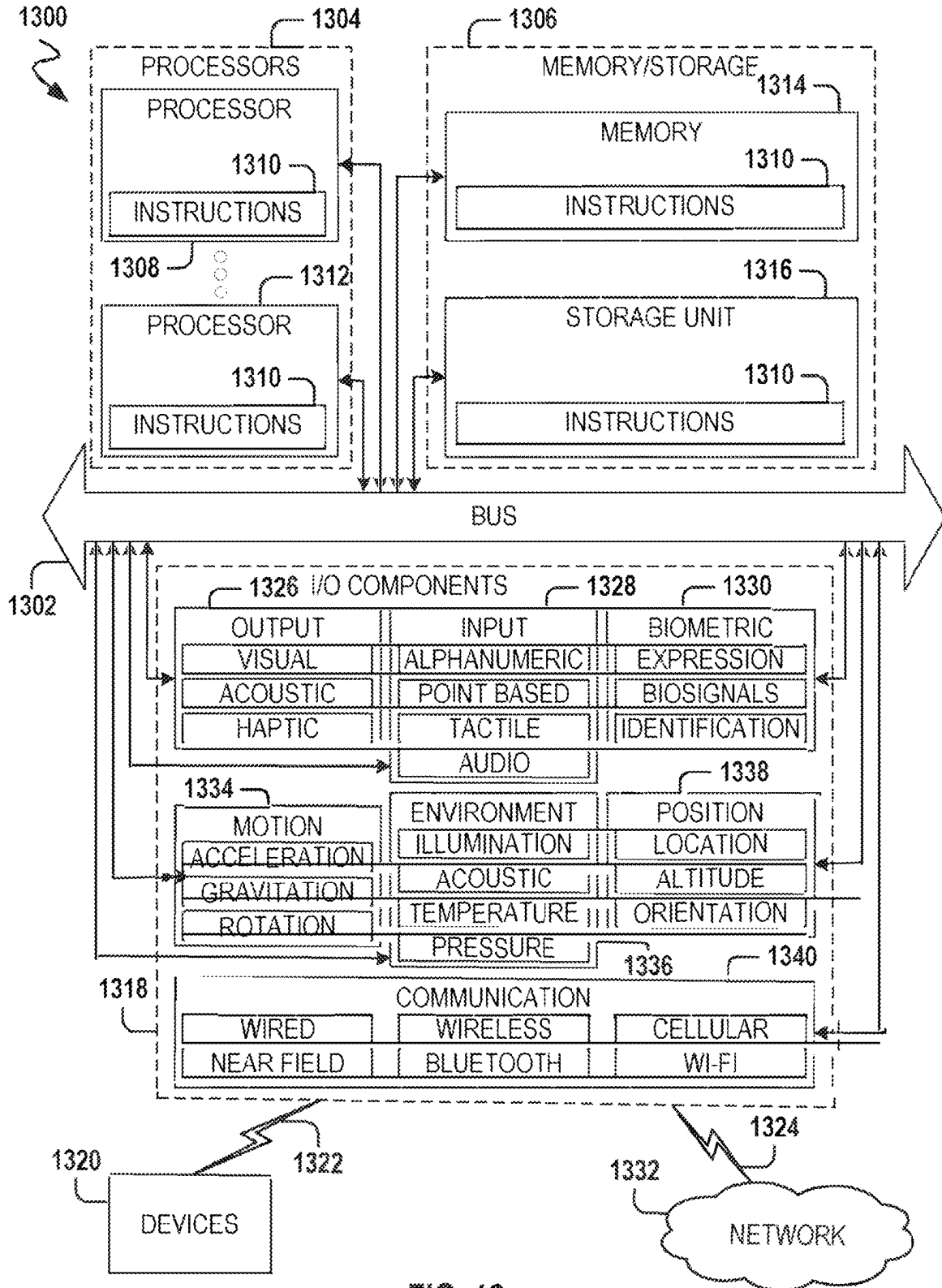


FIG. 13

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PARALLEL PREDICTION OF MULTIPLE IMAGE ASPECTS

RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 17/523,998 filed Nov. 11, 2021, which is a continuation of U.S. patent application Ser. No. 15/294,762, filed Oct. 16, 2016, now U.S. Pat. No. 11,200,273 issued Dec. 14, 2021, both of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

Embodiments of the present disclosure relate generally to image search.

BACKGROUND

Present techniques that analyze images to characterize aspects of the images rely on manual techniques that do not scale. Automated techniques use neural networks to characterize aspects of the images. However, because each aspect uses its own neural network, automated techniques also do not scale as the number of aspects increases.

BRIEF DESCRIPTION OF THE DRAWINGS

Various ones of the appended drawings merely illustrate example embodiments of the present disclosure and cannot be considered as limiting its scope.

FIG. 1 is a block diagram illustrating a networked system, according to some example embodiments.

FIG. 2 is a diagram illustrating the operation of the intelligent assistant, according to some example embodiments.

FIG. 3 illustrates the features of the artificial intelligence (AI) framework, according to some example embodiments.

FIG. 4 is a diagram illustrating a service architecture according to some example embodiments.

FIG. 5 is a block diagram for implementing the AI framework, according to some example embodiments.

FIG. 6 depicts a diagram of a category hierarchy tree that arranges each publications of a publication corpus into a hierarchy in accordance with some example embodiments.

FIG. 7 is an example process flow of training a machine learned model.

FIGS. 8-11 are example process flows of providing, in parallel, multiple aspect probabilities of an input image.

FIG. 12 is a diagram illustrating heat maps of location data of locations in the input image that have an aspect value.

FIG. 13 is a block diagram illustrating components of a machine according to some example embodiments.

The headings provided herein are merely for convenience and do not necessarily affect the scope or meaning of the terms used.

DETAILED DESCRIPTION

The description that follows describes systems, methods, techniques, instruction sequences, and computing machine program products that illustrate example embodiments of the present subject matter. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide an understanding of various embodiments of the present subject matter. It will be evi-

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dent, however, to those skilled in the art, that embodiments of the present subject matter may be practiced without some or other of these specific details. In general, well-known instruction instances, protocols, structures, and techniques have not been shown in detail. Examples merely typify possible variations. Unless explicitly stated otherwise, structures (e.g., structural components, such as modules) are optional and may be combined or subdivided, and operations (e.g., in a procedure, algorithm, or other function) may vary in sequence or be combined or subdivided.

Example embodiments that analyze images to characterize aspects of the images rely on a same neural network to characterize multiple aspects in parallel. Because additional neural networks are not required for additional aspects, such an approach scales with increased aspects.

FIG. 1 is a block diagram illustrating a networked system, according to some example embodiments. With reference to FIG. 1, an example embodiment of a high-level client-server-based network architecture 100 is shown. A networked system 102, in the example forms of a network-based marketplace or payment system, provides server-side functionality via a network 104 (e.g., the Internet or wide area network (WAN)) to one or more client devices 110. FIG. 1 illustrates, for example, a web client 112 (e.g., a browser, such as the Internet Explorer® browser developed by Microsoft® Corporation of Redmond, Washington State), an application 114, and a programmatic client 116 executing on client device 110.

The client device 110 may comprise, but are not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smart phones, tablets, ultra books, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may utilize to access the networked system 102. In some embodiments, the client device 110 may comprise a display module (not shown) to display information (e.g., in the form of user interfaces). In further embodiments, the client device 110 may comprise one or more of a touch screens, accelerometers, gyroscopes, cameras, microphones, global positioning system (GPS) devices, and so forth. The client device 110 may be a device of a user that is used to perform a transaction involving digital items within the networked system 102. In one embodiment, the networked system 102 is a network-based marketplace that responds to requests for product listings, publishes publications comprising item listings of products available on the network-based marketplace, and manages payments for these marketplace transactions. One or more users 106 may be a person, a machine, or other means of interacting with client device 110. In embodiments, the user 106 is not part of the network architecture 100, but may interact with the network architecture 100 via client device 110 or another means. For example, one or more portions of network 104 may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a cellular telephone network, a wireless network, a WiFi network, a WiMax network, another type of network, or a combination of two or more such networks.

Each of the client device 110 may include one or more applications (also referred to as “apps”) such as, but not limited to, a web browser, messaging application, electronic mail (email) application, an e-commerce site application

(also referred to as a marketplace application), and the like. In some embodiments, if the e-commerce site application is included in a given one of the client device **110**, then this application is configured to locally provide the user interface and at least some of the functionalities with the application configured to communicate with the networked system **102**, on an as needed basis, for data or processing capabilities not locally available (e.g., access to a database of items available for sale, to authenticate a user, to verify a method of payment, etc.). Conversely if the e-commerce site application is not included in the client device **110**, the client device **110** may use its web browser to access the e-commerce site (or a variant thereof) hosted on the networked system **102**.

One or more users **106** may be a person, a machine, or other means of interacting with the client device **110**. In example embodiments, the user **106** is not part of the network architecture **100**, but may interact with the network architecture **100** via the client device **110** or other means. For instance, the user provides input (e.g., touch screen input or alphanumeric input) to the client device **110** and the input is communicated to the networked system **102** via the network **104**. In this instance, the networked system **102**, in response to receiving the input from the user, communicates information to the client device **110** via the network **104** to be presented to the user. In this way, the user can interact with the networked system **102** using the client device **110**.

An application program interface (API) server **216** and a web server **218** are coupled to, and provide programmatic and web interfaces respectively to, one or more application servers **140**. The application server **140** host the intelligent personal assistant system **142**, which includes the artificial intelligence framework **144**, each of which may comprise one or more modules or applications and each of which may be embodied as hardware, software, firmware, or any combination thereof.

The application server **140** is, in turn, shown to be coupled to one or more database servers **226** that facilitate access to one or more information storage repositories or databases **226**. In an example embodiment, the databases **226** are storage devices that store information to be posted (e.g., publications or listings) to the publication system **242**. The databases **226** may also store digital item information in accordance with example embodiments.

Additionally, a third-party application **132**, executing on third-party servers **130**, is shown as having programmatic access to the networked system **102** via the programmatic interface provided by the API server **216**. For example, the third-party application **132**, utilizing information retrieved from the networked system **102**, supports one or more features or functions on a website hosted by the third party. The third-party website, for example, provides one or more promotional, marketplace, or payment functions that are supported by the relevant applications of the networked system **102**.

Further, while the client-server-based network architecture **100** shown in FIG. **1** employs a client-server architecture, the present inventive subject matter is of course not limited to such an architecture, and could equally well find application in a distributed, or peer-to-peer, architecture system, for example. The various publication system **142**, payment system **144**, and personalization system **150** could also be implemented as standalone software programs, which do not necessarily have networking capabilities.

The web client **212** may access the intelligent personal assistant system **142** via the web interface supported by the web server **218**. Similarly, the programmatic client **116** accesses the various services and functions provided by the

intelligent personal assistant system **142** via the programmatic interface provided by the API server **216**.

Additionally, a third-party application(s) **208**, executing on a third-party server(s) **130**, is shown as having programmatic access to the networked system **102** via the programmatic interface provided by the API server **114**. For example, the third-party application **208**, utilizing information retrieved from the networked system **102**, may support one or more features or functions on a website hosted by the third party. The third-party website may, for example, provide one or more promotional, marketplace, or payment functions that are supported by the relevant applications of the networked system **102**.

FIG. **2** is a diagram illustrating the operation of the intelligent assistant, according to some example embodiments. Today's online shopping is impersonal, unidirectional, and not conversational. Buyers cannot speak in plain language to convey their wishes, making it difficult to convey intent. Shopping on a commerce site is usually more difficult than speaking with a salesperson or a friend about a product, so oftentimes buyers have trouble finding the products they want.

Embodiments present a personal shopping assistant, also referred to as an intelligent assistant, that supports a two-way communication with the shopper to build context and understand the intent of the shopper, enabling delivery of better, personalized shopping results. The intelligent assistant has a natural, human-like dialog, that helps a buyer with ease, increasing the likelihood that the buyer will reuse the intelligent assistant for future purchases.

The artificial intelligence framework **144** understands the user and the available inventory to respond to natural-language queries and has the ability to deliver incremental improvements in anticipating and understanding the customer and their needs.

The artificial intelligence framework (AIF) **144** includes a dialogue manager **504**, natural language understanding (NLU) **206**, computer vision **208**, speech recognition **210**, search **218**, and orchestrator **220**. The AIF **144** is able to receive different kinds of inputs, such as text input **212**, image input **214** and voice input **216**, to generate relevant results **222**. As used herein, the AIF **144** includes a plurality of services (e.g., NLU **206**, computer vision **208**) that are implemented by corresponding servers, and the terms service or server may be utilized to identify the service and the corresponding service.

The natural language understanding (NLU) **206** unit processes natural language text input **212**, both formal and informal language, detects the intent of the text, and extracts useful information, such as objects of interest and their attributes. The natural language user input can thus be transformed into a structured query using rich information from additional knowledge to enrich the query even further. This information is then passed on to the dialog manager **504** through the orchestrator **220** for further actions with the user or with the other components in the overall system. The structured and enriched query is also consumed by search **218** for improved matching. The text input may be a query for a product, a refinement to a previous query, or other information to an object of relevance (e.g., shoe size).

The computer vision **208** takes image as an input and performs image recognition to identify the characteristics of the image (e.g., item the user wants to ship), which are then transferred to the NLU **206** for processing. The speech recognition **210** takes speech **216** as an input and performs language recognition to convert speech to text, which is then transferred to the NLU for processing.

The NLU **206** determines the object, the aspects associated with the object, how to create the search interface input, and how to generate the response. For example, the AI **144** may ask questions to the user to clarify what the user is looking for. This means that the AIF **144** not only generates results, but also may create a series of interactive operations to get to the optimal, or close to optimal, results **222**.

For example, in response to the query, "Can you find me a pair of red nike shoes?" the AIF **144** may generate the following parameters: <intent:shopping, statement-type: question, dominant-object:shoes, target:self, color:red, brand:nike>. To the query, "I am looking for a pair of sunglasses for my wife," the NLU may generate <intent: shopping, statement-type: statement, dominant-object: sunglasses, target:wife, target-gender:female>.

The dialogue manager **504** is the module that analyzes the query of a user to extract meaning, and determines if there is a question that needs to be asked in order to refine the query, before sending the query to search **218**. The dialogue manager **504** uses the current communication in context of the previous communication between the user and the artificial intelligence framework **144**. The questions are automatically generated dependent on the combination of the accumulated knowledge (e.g., provided by a knowledge graph) and what search can extract out of the inventory. The dialogue manager's job is to create a response for the user. For example, if the user says, "hello," the dialogue manager **504** generates a response, "Hi, my name is bot."

The orchestrator **220** coordinates the interactions between the other services within the artificial intelligence framework **144**. More details are provided below about the interactions of the orchestrator **220** with other services with reference to FIG. 5.

FIG. 3 illustrates the features of the artificial intelligence (AI) framework **144**, according to some example embodiments. The AIF **144** is able to interact with several input channels **304**, such as native commerce applications, chat applications, social networks, browsers, etc. In addition, the AIF **144** understands the intent **306** expressed by the user. For example, the intent may include a user looking for a good deal, or a user looking for a gift, or a user on a mission to buy a specific product, a user looking for suggestions, etc.

Further, the AIF **144** performs proactive data extraction **310** from multiple sources, such as social networks, email, calendar, news, market trends, etc. The AIF **144** knows about user details **312**, such as user preferences, desired price ranges, sizes, affinities, etc. The AIF **144** facilitates a plurality of services within the service network, such as product search, personalization, recommendations, checkout features, etc. The output **308** may include recommendations, results, etc.

The AIF **144** is an intelligent and friendly system that understands the user's intent (e.g., targeted search, compare, shop, browse), mandatory parameters (e.g., product, product category, item), optional parameters (e.g., aspects of the item, color, size, occasion), as well as implicit information (e.g., geo location, personal preferences, age, gender). The AIF **144** responds with a well designed response in plain language.

For example, the AIF **144** may process inputs queries, such as: "Hey! Can you help me find a pair of light pink shoes for my girlfriend please? With heels. Up to \$200. Thanks;" "I recently searched for a men's leather jacket with a classic James Dean look. Think almost Harrison Ford's in the new Star Wars movie. However, I'm looking for quality

in a price range of \$200-300. Might not be possible, but I wanted to see!"; or "I'm looking for a black Northface Thermoball jacket."

Instead of a hardcoded system, the AIF **144** provides a configurable, flexible interface with machine learning capabilities for ongoing improvement. The AIF **144** supports a commerce system that provides value (connecting the user to the things that the user wants), intelligence (knowing and learning from the user and the user behavior to recommend the right items), convenience (offering a plurality of user interfaces), easy of-use, and efficiency (saves the user time and money).

FIG. 4 is a diagram illustrating a service architecture **400** according to some embodiments. The service architecture **400** presents various views of the service architecture in order to describe how the service architecture may be deployed on various data centers or cloud services. The architecture **400** represents a suitable environment for implementation of the embodiments described herein.

The service architecture **402** represents how a cloud architecture typically appears to a user, developer and so forth. The architecture is generally an abstracted representation of the actual underlying architecture implementation, represented in the other views of FIG. 1. For example, the service architecture **402** comprises a plurality of layers, that represent different functionality and/or services associated with the service architecture **402**.

The experience service layer **404** represents a logical grouping of services and features from the end customer's point of view, built across different client platforms, such as applications running on a platform (mobile phone, desktop, etc.), web based presentation (mobile web, desktop web browser, etc.), and so forth. It includes rendering user interfaces and providing information to the client platform so that appropriate user interfaces can be rendered, capturing client input, and so forth. In the context of a marketplace, examples of services that would reside in this layer are home page (e.g., home view), view item listing, search/view search results, shopping cart, buying user interface and related services, selling user interface and related services, after sale experiences (posting a transaction, feedback, etc.), and so forth. In the context of other systems, the experience service layer **404** would incorporate those end user services and experiences that are embodied by the system.

The API layer **406** contains APIs which allow interaction with business process and core layers. This allows third party development against the service architecture **402** and allows third parties to develop additional services on top of the service architecture **402**.

The business process service layer **408** is where the business logic resides for the services provided. In the context of a marketplace this is where services such as user registration, user sign in, listing creation and publication, add to shopping cart, place an offer, checkout, send invoice, print labels, ship item, return item, and so forth would be implemented. The business process service layer **408** also orchestrates between various business logic and data entities and thus represents a composition of shared services. The business processes in this layer can also support multi-tenancy in order to increase compatibility with some cloud service architectures.

The data entity service layer **410** enforces isolation around direct data access and contains the services upon which higher level layers depend. Thus, in the marketplace context this layer can comprise underlying services like order management, financial institution management, user

account services, and so forth. The services in this layer typically support multi-tenancy.

The infrastructure service layer **412** comprises those services that are not specific to the type of service architecture being implemented. Thus, in the context of a marketplace, the services in this layer are services that are not specific or unique to a marketplace. Thus, functions like cryptographic functions, key management, CAPTCHA, authentication and authorization, configuration management, logging, tracking, documentation and management, and so forth reside in this layer.

Embodiments of the present disclosure will typically be implemented in one or more of these layers. In particular, the AIF **144**, as well as the orchestrator **220** and the other services of the AIF **144**.

The data center **414** is a representation of the various resource pools **416** along with their constituent scale units. This data center representation illustrates the scaling and elasticity that comes with implementing the service architecture **402** in a cloud computing model. The resource pool **416** is comprised of server (or compute) scale units **420**, network scale units **418** and storage scale units **422**. A scale unit is a server, network and/or storage unit that is the smallest unit capable of deployment within the data center. The scale units allow for more capacity to be deployed or removed as the need increases or decreases.

The network scale unit **418** contains one or more networks (such as network interface units, etc.) that can be deployed. The networks can include, for example virtual LANs. The compute scale unit **420** typically comprise a unit (server, etc.) that contains a plurality processing units, such as processors. The storage scale unit **422** contains one or more storage devices such as disks, storage attached networks (SAN), network attached storage (NAS) devices, and so forth. These are collectively illustrated as SANs in the description below. Each SAN may comprise one or more volumes, disks, and so forth.

The remaining view of FIG. 1 illustrates another example of a service architecture **400**. This view is more hardware focused and illustrates the resources underlying the more logical architecture in the other views of FIG. 1. A cloud computing architecture typically has a plurality of servers or other systems **424**, **426**. These servers comprise a plurality of real and/or virtual servers. Thus the server **424** comprises server **1** along with virtual servers **1A**, **1B**, **1C** and so forth.

The servers are connected to and/or interconnected by one or more networks such as network A **428** and/or network B **430**. The servers are also connected to a plurality of storage devices, such as SAN 1 (**436**), SAN 2 (**438**) and so forth. SANs are typically connected to the servers through a network such as SAN access A **432** and/or SAN access B **434**.

The compute scale units **420** are typically some aspect of servers **424** and/or **426**, like processors and other hardware associated therewith. The network scale units **418** typically include, or at least utilize the illustrated networks A (**428**) and B (**432**). The storage scale units typically include some aspect of SAN 1 (**436**) and/or SAN 2 (**438**). Thus, the logical service architecture **402** can be mapped to the physical architecture.

Services and other implementation of the embodiments described herein will run on the servers or virtual servers and utilize the various hardware resources to implement the disclosed embodiments.

FIG. 5 is a block diagram for implement the AIF **144**, according to some example embodiments. Specifically, the intelligent personal assistant system **106** of FIG. 2 is shown

to include a front end component **502** (FE) by which the intelligent personal assistant system **106** communicates (e.g., over the network **104**) with other systems within the network architecture **100**. The front end component **502** can communicate with the fabric of existing messaging systems. As used herein, the term messaging fabric refers to a collection of APIs and services that can power third party platforms such as Facebook messenger, Microsoft Cortana, and others “bots.” In one example, a messaging fabric can support an online commerce ecosystem that allows users to interact with commercial intent. Output of the front end component **502** can be rendered in a display of a client device, such as the client device **110** in FIG. 1 as part of an interface with the intelligent personal assistant.

The front end component **502** of the intelligent personal assistant system **106** is coupled to a back end component **504** for the front end (BFF) that operates to link the front end component **502** with the AIF **144**. The artificial intelligence framework **144** includes several components discussed below.

In one example embodiment, an orchestrator **220** orchestrates communication of components inside and outside the artificial intelligence framework **144**. Input modalities for the AI orchestrator **206** are derived from a computer vision component **208**, a speech recognition component **210**, and a text normalization component which may form part of the speech recognition component **210**. The computer vision component **208** may identify objects and attributes from visual input (e.g., photo). The speech recognition component **210** converts audio signals (e.g., spoken utterances) into text. The text normalization component operates to make input normalization, such as language normalization by rendering emoticons into text, for example. Other normalization is possible such as orthographic normalization, foreign language normalization, conversational text normalization, and so forth.

The artificial intelligence framework **144** further includes a natural language understanding (NLU) component **206** that operates to parse and extract user intent and intent parameters (for example mandatory or optional parameters). The NLU component **206** is shown to include sub-components such as a spelling corrector (speller), a parser, a named entity recognition (NER) sub-component, a knowledge graph, and a word sense detector (WSD).

The artificial intelligence framework **144** further includes a dialog manager **204** that operates to understand a “completeness of specificity” (for example of an input, such as a search query or utterance) and decide on a next action type and a parameter (e.g., “search” or “request further information from user”). In one example, the dialog manager **204** operates in association with a context manager **518** and a natural language generation (NLG) component **512**. The context manager **518** manages the context and communication of a user with respect to online personal assistant (or “bot”) and the assistant’s associated artificial intelligence. The context manager **518** comprises two parts: long term history and short term memory. Data entries into one or both of these parts can include the relevant intent and all parameters and all related results of a given input, bot interaction, or turn of communication, for example. The NLG component **512** operates to compose a natural language utterance out of a AI message to present to a user interacting with the intelligent bot.

A search component **218** is also included within the artificial intelligence framework **144**. As shown, the search component **218** has a front-end and a back-end unit. The back-end unit operates to manage item and product inven-

tory and provide functions of searching against the inventory, optimizing towards a specific tuple of intent and intent parameters. An identity service **522** component, that may or may not form part of artificial intelligence framework **144**, operates to manage user profiles, for example explicit information in the form of user attributes (e.g., “name,” “age,” “gender,” “geolocation”), but also implicit information in forms such as “information distillates” such as “user interest,” or “similar persona,” and so forth. The identity service **522** includes a set of policies, APIs, and services that elegantly centralizes all user information, enabling the AIF **144** to have insights into the users’ wishes. Further, the identity service **522** protects the commerce system and its users from fraud or malicious use of private information.

The functionalities of the artificial intelligence framework **144** can be set into multiple parts, for example decision-making and context parts. In one example, the decision-making part includes operations by the orchestrator **220**, the NLU component **206** and its subcomponents, the dialog manager **204**, the NLG component **512**, the computer vision component **208** and speech recognition component **210**. The context part of the AI functionality relates to the parameters (implicit and explicit) around a user and the communicated intent (for example, towards a given inventory, or otherwise). In order to measure and improve AI quality over time, in some example embodiments, the artificial intelligence framework **144** is trained using sample queries (e.g., a development set) and tested on a different set of queries (e.g., an evaluation set), both sets to be developed by human curation or from use data. Also, the artificial intelligence framework **144** is to be trained on transaction and interaction flows defined by experienced curation specialists, or human override **524**. The flows and the logic encoded within the various components of the artificial intelligence framework **144** define what follow-up utterance or presentation (e.g., question, result set) is made by the intelligent assistant based on an identified user intent.

The intelligent personal assistant system **106** seeks to understand a user’s intent (e.g., targeted search, compare, shop, browse, and so forth), mandatory parameters (e.g., product, product category, item, and so forth), and optional parameters (e.g., explicit information, e.g., aspects of item/product, occasion, and so forth), as well as implicit information (e.g., geolocation, personal preferences, age and gender, and so forth) and respond to the user with a content-rich and intelligent response. Explicit input modalities can include text, speech, and visual input and can be enriched with implicit knowledge of user (e.g., geolocation, gender, birthplace, previous browse history, and so forth). Output modalities can include text (such as speech, or natural language sentences, or product-relevant information, and images on the screen of a smart device e.g., client device **110**). Input modalities thus refer to the different ways users can communicate with the bot. Input modalities can also include keyboard or mouse navigation, touch-sensitive gestures, and so forth.

In relation to a modality for the computer vision component **208**, a photograph can often represent what a user is looking for better than text. Also, the computer vision component **208** may be used to form shipping parameters based on the image of the item to be shipped. The user may not know what an item is called, or it may be hard or even impossible to use text for fine detailed information that an expert may know, for example a complicated pattern in apparel or a certain style in furniture. Moreover, it is inconvenient to type complex text queries on mobile phones and long text queries typically have poor recall. Key func-

tionalties of the computer vision component **208** include object localization, object recognition, optical character recognition (OCR) and matching against inventory based on visual cues from an image or video. A bot enabled with computer vision is advantageous when running on a mobile device which has a built-in camera. Powerful deep neural networks can be used to enable computer vision applications.

With reference to the speech recognition component **210**, a feature extraction component operates to convert raw audio waveform to some-dimensional vector of numbers that represents the sound. This component uses deep learning to project the raw signal into a high-dimensional semantic space. An acoustic model component operates to host a statistical model of speech units, such as phonemes and allophones. These can include Gaussian Mixture Models (GMM) although the use of Deep Neural Networks is possible. A language model component uses statistical models of grammar to define how words are put together in a sentence. Such models can include n-gram-based models or Deep Neural Networks built on top of word embeddings. A speech-to-text (STT) decoder component converts a speech utterance into a sequence of words typically leveraging features derived from a raw signal using the feature extraction component, the acoustic model component, and the language model component in a Hidden Markov Model (HMM) framework to derive word sequences from feature sequences. In one example, a speech-to-text service in the cloud has these components deployed in a cloud framework with an API that allows audio samples to be posted for speech utterances and to retrieve the corresponding word sequence. Control parameters are available to customize or influence the speech-to-text process.

Machine-learning algorithms may be used for matching, relevance, and final re-ranking by the AIF **144** services. Machine learning is a field of study that gives computers the ability to learn without being explicitly programmed. Machine learning explores the study and construction of algorithms that can learn from and make predictions on data. Such machine-learning algorithms operate by building a model from example inputs in order to make data-driven predictions or decisions expressed as outputs. Machine-learning algorithms may also be used to teach how to implement a process.

Deep learning models, deep neural network (DNN), recurrent neural network (RNN), convolutional neural network (CNN), and long short-term CNN, as well as other ML models and IR models may be used. For example, search **218** may use n-gram, entity, and semantic vector-based query to product matching. Deep-learned semantic vectors give the ability to match products to non-text inputs directly. Multi-leveled relevance filtration may use BM25, predicted query leaf category+product leaf category, semantic vector similarity between query and product, and other models, to pick the top candidate products for the final re-ranking algorithm.

Predicted click-through-rate and conversion rate, as well as GMV, constitutes the final re-ranking formula to tweak functionality towards specific business goals, more shopping engagement, more products purchased, or more GMV. Both the click prediction and conversion prediction models take in query, user, seller and product as input signals. User profiles are enriched by learning from onboarding, sideboarding, and user behaviors to enhance the precision of the models used by each of the matching, relevance, and ranking

stages for individual users. To increase the velocity of model improvement, offline evaluation pipeline is used before online A/B testing.

In one example of an artificial intelligence framework 144, two additional parts for the speech recognition component 210 are provided, a speaker adaptation component and an LM adaptation component. The speaker adaptation component allows clients of an STT system (e.g., speech recognition component 210) to customize the feature extraction component and the acoustic model component for each speaker. This can be important because most speech-to-text systems are trained on data from a representative set of speakers from a target region and typically the accuracy of the system depends heavily on how well the target speaker matches the speakers in the training pool. The speaker adaptation component allows the speech recognition component 210 (and consequently the artificial intelligence framework 144) to be robust to speaker variations by continuously learning the idiosyncrasies of a user's intonation, pronunciation, accent, and other speech factors and apply these to the speech-dependent components, e.g., the feature extraction component, and the acoustic model component. While this approach utilizes a non-significant-sized voice profile to be created and persisted for each speaker, the potential benefits of accuracy generally far outweigh the storage drawbacks.

The language model (LM) adaptation component operates to customize the language model component and the speech-to-text vocabulary with new words and representative sentences from a target domain, for example, inventory categories or user personas. This capability allows the artificial intelligence framework 144 to be scalable as new categories and personas are supported.

The AIF's goal is to provide a scalable and expandable framework for AI, one in which new activities, also referred to herein as missions, can be accomplished dynamically using the services that perform specific natural-language processing functions. Adding a new service does not require to redesign the complete system. Instead, the services are prepared (e.g., using machine-learning algorithms) if necessary, and the orchestrator is configured with a new sequence related to the new activity. More details regarding the configuration of sequences are provided below with reference to other figures and associated text.

Embodiments presented herein provide for dynamic configuration of the orchestrator 220 to learn new intents and how to respond to the new intents. In some example embodiments, the orchestrator 220 "learns" new skills by receiving a configuration for a new sequence associated with the new activity. The sequence specification includes a sequence of interactions between the orchestrator 220 and a set of one or more service servers from the AIF 144. In some example embodiments, each interaction of the sequence includes (at least): identification for a service server, a call parameter definition to be passed with a call to the identified service server; and a response parameter definition to be returned by the identified service server.

In some example embodiments, the services within the AIF 144, except for the orchestrator 220, are not aware of each other, e.g., they do not interact directly with each other. The orchestrator 220 manages all the interactions with the other servers. Having the central coordinating resource simplifies the implementation of the other services, which need not be aware of the interfaces (e.g., APIs) provided by the other services. Of course, there can be some cases where a direct interface may be supported between pairs of services.

FIG. 6 depicts a diagram of a category hierarchy tree that arranges each publications of a publication corpus into a hierarchy in accordance with some example embodiments. In some example embodiments, the publication categories are then organized into a hierarchy (e.g., a map or tree), such that more general categories include more specific categories. Each node in the tree or map is a publication category that has a parent category (e.g., a more general category with which the publication category is associated) and potentially one or more child categories (e.g., narrow or more specific categories associated with the publication category.). Each publication category is associated with a particular static webpage.

In accordance with some example embodiments, a plurality of publication is grouped together into publication categories. In this example, each category is labeled with a letter (e.g., category A-category AJ). In addition, every publication category is organized as part of a hierarchy of categories.

In this example, category A is a general product category that all other publication categories descend from. Publications in category A are then divided in to at least two different publication categories, category B and category C. It should be noted that each parent category (e.g., in this case category A is a parent category to both Category B and Category C) may include a large number of child categories (e.g., subcategories).

In this example, publication categories B and C both have subcategories (or child categories). For example, if Category A is clothing publications, Category B can be Men's clothes publications and Category C is Women's clothes publications. Subcategories for Category B include category D, category E, and category F. Each of subcategories D, E, and F have a different number of subcategories, depending on the specific details of the publications covered by each subcategory.

For example, if category D is active wear publications, category E is formal wear publications, and category F is outdoor wear publications, each subcategory includes different numbers and types of subcategories. For example, category D (active wear publications in this example) includes subcategories I and J. Subcategory I includes Active Footwear publications (for this example) and Subcategory J includes t-shirt publications. As a result of the differences between these two subcategories, subcategory I includes four additional subcategories (subcategories K-N) to represent different types of active footwear publications (e.g., running shoe publications, basketball shoe publications, climbing shoe publications, and tennis shoe publications). In contrast, subcategory J (which, in this example, is for t-shirt publications) does not include any subcategories (although in a real product database a t-shirt publications category would likely include subcategories).

Thus, each category has a parent category (except for the uppermost product category) which represents a more general category of publications and one or more child categories or subcategories (which are a more specific publications category within the more general category). Thus, category E has two sub-categories, O and P, and each subcategory has two child product categories, categories Q and R and categories S and T, respectively. Similarly, category F has three sub-categories (U, V, and W).

Category C, a product category that has Category A as its parent, includes two additional subcategories (G and H). Category G includes two children (X and AF). Category X includes subcategories Y and Z, and Y includes AA-AE.

Category H includes subcategories AG and AH. Category AG includes categories AI and AJ.

Aspects are characteristics of publications listed in a given category. Aspects are specific to a specific category. Aspects of different categories may or may not overlap, or be unique to a specific category.

FIG. 7 is an example process flow of training a machine learned model. At 710, a training image is input to a machine learned model. At 720, the training image is processed with the machine learned model. At 730, multiple training aspect values are output in parallel from the machine learned model. At 740, the machine learned model is trained by feeding back to the machine learned model whether or not each of the multiple training aspect values was correct.

In an example embodiment, a machine-learned model is used to embed the deep latent semantic meaning of a given listing title and project it to a shared semantic vector space. A vector space can be referred to as a collection of objects called vectors. Vectors spaces can be characterized by their dimension, which specifies the number of independent directions in the space. A semantic vector space can represent phrases and sentences and can capture semantics for image search and image characterization tasks. In further embodiments, a semantic vector space can represent audio sounds, phrases, or music; video clips; and images and can capture semantics for image search and image characterization tasks.

In various embodiments, machine learning is used to maximize the similarity between the source (X), for example, a listing title, and the target (Y), the search query. A machine-learned model may be based on deep neural networks (DNN) or convolutional neural networks (CNN). The DNN is an artificial neural network with multiple hidden layers of units between the input and output layers. The DNN can apply the deep learning architecture to recurrent neural networks. The CNN is composed of one or more convolution layers with fully connected layers (such as those matching a typical artificial neural network) on top. The CNN also uses tied weights and pooling layers. Both the DNN and CNN can be trained with a standard backpropagation algorithm.

When a machine-learned model is applied to mapping a specific <source, target> pair, the parameters for machine-learned Source Model and machine-learned Target Model are optimized so that relevant <source, target> pair has closer vector representation distance. The following formula can be used to compute the minimum distance.

$$\text{SrcMod}^*, \text{TgtMod}^* = \underset{\sum_k \text{ in all training pairs}}{\operatorname{argmin}} \|\text{SrcVec}^k - \text{TgtVec}^k\|$$

Where,

SrcSeq=a source sequence;

TgtSeq=a target sequence;

SrcMod=source machine-learned model;

TgtMod=target machine-learned model;

SrcVec=a continuous vector representation for a source sequence (also referred to the semantic vector of the source); and

TgtVec=a continuous vector representation for a target sequence (also referred to as semantic vector of the target).

The source machine-learned model encodes the source sequence into a continuous vector representation. The target machine-learned model encodes the target sequence into a continuous vector representation. In an example embodiment, the vectors each have approximately 100 dimensions.

In other embodiments, any number of dimensions may be used. In example embodiments, the dimensions of the semantic vectors are stored in a KD tree structure. The KD tree structure can be referred to as a space-partitioning data structure for organizing points in a KD space. The KD tree can be used to perform the nearest-neighbor lookup. Thus, given a source point in space, the nearest-neighbor lookup may be used to identify the closest point to the source point.

FIGS. 8-11 are example process flows of providing, in parallel, multiple aspect probabilities of an input image. In FIG. 8, at 810 an input image is transmitted from a device operated by a user. The user may be searching for a publication in a publication corpus. The user may be posting a new publication with publication images, and rely on the process flow to help provide the category. At 820, an input semantic vector corresponding to the input image is accessed. At this point, the process flow splits. At 830, the input semantic vector and publication image vectors are converted into binary representations. At 840, closest matches are identified between the input semantic vector and publication image vectors that are representative of multiple aspects. The machine learned model is used along with XOR operations for speed. A number of common bits from the XOR operation is a measure of similarity. In an alternative flow, at 850 closest matches are identified between the input semantic vector and publication image vectors that are representative of multiple aspects by finding nearest neighbors in semantic vector space. After either of the previous split process flows, at 870 the multiple aspect probabilities are provided, based on the machine learned model.

The process flow of FIG. 9 is generally similar to FIG. 8. At 910, the input image is missing category metadata. At 970, at least one aspect probability is provided for the input image that was missing metadata of the aspect value. In another embodiment, at least one aspect probability is provided for the input image that was not missing metadata, to double check the metadata.

The process flow of FIG. 10 is generally similar to FIG. 9. At 1070, the missing aspect metadata is added to the input image, responsive to an aspect probability exceeding a minimum threshold.

The process flow of FIG. 11 is generally similar to FIG. 10. At 1170, location metadata is added to the input image, such that the location metadata has the location in the input image that has the aspect value.

FIG. 12 is a diagram illustrating heat maps of location data of locations in the input image that have an aspect value. Images 1210, 1220, and 1230 are input images having the aspect value of hoodie. Respective heat maps 1212, 1222, and 1232 are heat maps having the locations in the images that have the aspect value of hoodie. For example, the hot regions of the heat maps correspond to regions of the input image that have parts of a hoodie, such as a hood, sleeves, and pockets.

FIG. 13 is a block diagram illustrating components of a machine 1300, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. 13 shows a diagrammatic representation of the machine 1300 in the example form of a computer system, within which instructions 1310 (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine 1300 to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions 1310 may cause the machine 1300 to execute the flow diagrams of other Figures.

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Additionally, or alternatively, the instructions **1310** may implement the servers associated with the services and components of other Figures, and so forth. The instructions **1310** transform the general, non-programmed machine **1300** into a particular machine **1300** programmed to carry out the described and illustrated functions in the manner described.

In alternative embodiments, the machine **1300** operates as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1300** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1300** may comprise, but not be limited to, a switch, a controller, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1310**, sequentially or otherwise, that specify actions to be taken by the machine **1300**. Further, while only a single machine **1300** is illustrated, the term “machine” shall also be taken to include a collection of machines **1300** that individually or jointly execute the instructions **1310** to perform any one or more of the methodologies discussed herein.

The machine **1300** may include processors **1304**, memory/storage **1306**, and I/O components **1318**, which may be configured to communicate with each other such as via a bus **1302**. In an example embodiment, the processors **1304** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) processor, a Complex Instruction Set Computing (CISC) processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **1308** and a processor **1312** that may execute the instructions **1310**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. **13** shows multiple processors **1304**, the machine **1300** may include a single processor with a single core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

The memory/storage **1306** may include a memory **1314**, such as a main memory, or other memory storage, and a storage unit **1313**, both accessible to the processors **1304** such as via the bus **1302**. The storage unit **1313** and memory **1314** store the instructions **1310** embodying any one or more of the methodologies or functions described herein. The instructions **1310** may also reside, completely or partially, within the memory **1314**, within the storage unit **1313**, within at least one of the processors **1304** (e.g., within the processor’s cache memory), or any suitable combination thereof, during execution thereof by the machine **1300**. Accordingly, the memory **1314**, the storage unit **1313**, and the memory of the processors **1304** are examples of machine-readable media.

As used herein, “machine-readable medium” means a device able to store instructions and data temporarily or permanently and may include, but is not limited to, random-

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access memory (RAM), read-only memory (ROM), buffer memory, flash memory, optical media, magnetic media, cache memory, other types of storage (e.g., Erasable Programmable Read-Only Memory (EEPROM)), and/or any suitable combination thereof. The term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store the instructions **1310**. The term “machine-readable medium” shall also be taken to include any medium, or combination of multiple media, that is capable of storing instructions (e.g., instructions **1310**) for execution by a machine (e.g., machine **1300**), such that the instructions, when executed by one or more processors of the machine (e.g., processors **1304**), cause the machine to perform any one or more of the methodologies described herein. Accordingly, a “machine-readable medium” refers to a single storage apparatus or device, as well as “cloud-based” storage systems or storage networks that include multiple storage apparatus or devices. The term “machine-readable medium” excludes signals per se.

The I/O components **1318** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1318** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones will likely include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1318** may include many other components that are not shown in FIG. **13**. The I/O components **1318** are grouped according to functionality merely for simplifying the following discussion, and the grouping is in no way limiting. In various example embodiments, the I/O components **1318** may include output components **1326** and input components **1328**. The output components **1326** may include visual components (e.g., a display such as a plasma display panel (PDP), a light emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The input components **1328** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and/or force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

In further example embodiments, the I/O components **1318** may include biometric components **1330**, motion components **1334**, environmental components **1336**, or position components **1338** among a wide array of other components. For example, the biometric components **1330** may include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components **1334** may include acceleration sensor components (e.g., accelerometer), gravitation sensor

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components, rotation sensor components (e.g., gyroscope), and so forth. The environmental components **1336** may include, for example, illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detect concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **1338** may include location sensor components (e.g., a Global Position System (GPS) receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

Communication may be implemented using a wide variety of technologies. The I/O components **1318** may include communication components **1340** operable to couple the machine **1300** to a network **1332** or devices **1320** via a coupling **1324** and a coupling **1322**, respectively. For example, the communication components **1340** may include a network interface component or other suitable device to interface with the network **1332**. In further examples, the communication components **1340** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1320** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

Moreover, the communication components **1340** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1340** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1340**, such as location via Internet Protocol (IP) geo-location, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

In various example embodiments, one or more portions of the network **1332** may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, the network **1332** or a portion of the network **1332** may include a wireless or cellular network and the coupling **1324** may be a Code

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Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or another type of cellular or wireless coupling. In this example, the coupling **1324** may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1xRTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long range protocols, or other data transfer technology.

The instructions **1310** may be transmitted or received over the network **1332** using a transmission medium via a network interface device (e.g., a network interface component included in the communication components **1340**) and utilizing any one of a number of well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **1310** may be transmitted or received using a transmission medium via the coupling **1322** (e.g., a peer-to-peer coupling) to the devices **1320**. The term “transmission medium” shall be taken to include any intangible medium that is capable of storing, encoding, or carrying the instructions **1310** for execution by the machine **1300**, and includes digital or analog communications signals or other intangible media to facilitate communication of such software.

Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component. Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements fall within the scope of the subject matter herein.

The embodiments illustrated herein are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed. Other embodiments may be used and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. The Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

As used herein, the term “or” may be construed in either an inclusive or exclusive sense. Moreover, plural instances may be provided for resources, operations, or structures described herein as a single instance. Additionally, boundaries between various resources, operations, modules, engines, and data stores are somewhat arbitrary, and particular operations are illustrated in a context of specific illustrative configurations. Other allocations of functionality are envisioned and may fall within a scope of various embodiments of the present disclosure. In general, structures and functionality presented as separate resources in the example configurations may be implemented as a combined

structure or resource. Similarly, structures and functionality presented as a single resource may be implemented as separate resources. These and other variations, modifications, additions, and improvements fall within a scope of embodiments of the present disclosure as represented by the appended claims. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense.

What is claimed is:

1. A method comprising:
 - receiving, by a network-based marketplace from a user device, a search query for an item listing of the network-based marketplace, the search query including an input image depicting an item;
 - identifying, by a machine learning system, matches between an input semantic vector representing the input image and one or more image vectors of one or more images from a publication corpus;
 - identifying a probability that the input image is associated with an aspect value that characterizes the item depicted in the input image based at least in part on the one or more image vectors;
 - adding metadata comprising the aspect value to the input image responsive to the probability that the input image is associated with the aspect value exceeding a threshold; and
 - transmitting, to the user device, a set of one or more search results responsive to the search query, wherein the set of one or more search results are identified based at least in part on the metadata and the input image.
2. The method of claim 1, further comprising:
 - receiving one or more aspect probabilities from the machine learning system, the one or more aspect probabilities identifying a respective probability that the input image has a respective aspect value of one or more aspect values characterizing the input image, the one or more aspect values comprising the aspect value.
3. The method of claim 2, wherein receiving, from the machine learning system, the one or more aspect probabilities comprises:
 - receiving the one or more aspect probabilities as parallel outputs from the machine learning system, wherein the parallel outputs include output location data with a location within a respective image.
4. The method of claim 1, further comprising:
 - determining that one or more aspect values characterizing the input image are missing, the one or more aspect values comprising a category of the item depicted in the input image, a color of the item depicted in the input image, or both, wherein identifying matches between the input semantic vector representing the input image and the one or more image vectors is based at least in part on the determining.
5. The method of claim 4, further comprising:
 - generating the input semantic vector based at least in part on the determining.
6. The method of claim 1, wherein the matches are identified based on a measure of similarity between the input semantic vector and the one or more image vectors.
7. The method of claim 1, further comprising:
 - ranking search results of the set of one or more search results based at least in part on the aspect value.
8. The method of claim 1, further comprising:
 - determining a product category for the item based on the aspect value, wherein the set of one or more search results are based on the product category.

9. A system comprising:
 - one or more hardware processors; and
 - a memory storing instructions that, when executed by the one or more hardware processors, cause the system to perform operations comprising:
 - receiving, from a user device, a search query for an item listing of a network-based marketplace, the search query including an input image depicting an item;
 - identifying, by a machine learning system, matches between an input semantic vector representing the input image and one or more image vectors of one or more images from a publication corpus;
 - identifying a probability that the input image is associated with an aspect value that characterizes the item depicted in the input image based at least in part on the one or more image vectors;
 - adding metadata comprising the aspect value to the input image responsive to the probability that the input image is associated with the aspect value exceeding a threshold; and
 - transmitting, to the user device, a set of one or more search results responsive to the search query, wherein the set of one or more search results are identified based at least in part on the metadata and the input image.
10. The system of claim 9, the operations further comprising:
 - receiving one or more aspect probabilities from the machine learning system, the one or more aspect probabilities identifying a respective probability that the input image has a respective aspect value of one or more aspect values characterizing the input image, the one or more aspect values comprising the aspect value.
11. The system of claim 10, wherein the instructions for receiving the one or more aspect probabilities, when executed by the one or more hardware processors, further cause the system to perform operations comprising receiving the one or more aspect probabilities as parallel outputs from the machine learning system, wherein the parallel outputs include output location data with a location within a respective image.
12. The system of claim 9, the operations further comprising:
 - determining that one or more aspect values characterizing the input image are missing, the one or more aspect values comprising a category of the item depicted in the input image, a color of the item depicted in the input image, or both, wherein identifying matches between the input semantic vector representing the input image and the one or more image vectors is based at least in part on the determining.
13. The system of claim 12, the operations further comprising:
 - generating the input semantic vector based at least in part on the determining.
14. The system of claim 9, wherein the matches are identified based on a measure of similarity between the input semantic vector and the one or more image vectors.
15. The system of claim 9, the operations further comprising:
 - ranking search results of the set of one or more search results based at least in part on the aspect value.
16. The system of claim 9, the operations further comprising:
 - determining a product category for the item based on the aspect value, wherein the set of one or more search results are based on the product category.

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17. A non-transitory computer readable storage medium comprising instructions that when executed by one or more processors cause a system to perform operations comprising:
 receiving, from a user device, a search query for an item
 listing of a network- based marketplace, the search 5
 query including an input image depicting an item;
 identifying, by a machine learning system, matches
 between an input semantic vector representing the input
 image and one or more image vectors of one or more
 images from a publication corpus;
 identifying a probability that the input image is associated
 with an aspect value that characterizes the item
 depicted in the input image based at least in part on the
 one or more image vectors;
 adding metadata comprising the aspect value to the input 15
 image responsive to the probability that the input image
 is associated with the aspect value exceeding a thresh-
 old; and

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transmitting, to the user device, a set of one or more
 search results responsive to the search query, wherein
 the set of one or more search results are identified based
 at least in part on the aspect value metadata and the
 input image.
 18. The non-transitory computer readable storage medium
 of claim 17, the operations further comprising:
 receiving one or more aspect probabilities from the
 machine learning system, the one or more aspect prob-
 abilities identifying a respective probability that the
 input image has a respective aspect value of one or
 more aspect values characterizing the input image, the
 one or more aspect values comprising the aspect value.
 19. The method of claim 1, wherein the probability
 indicates whether the input image includes the aspect value
 that characterizes the item depicted in the input image.

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